Technical Report - FastSpeech 2 (English and Telugu)

Status: draft, all sections written. Every section uses measured values recorded in `CHANGES.md`. Section 7 presents English and Telugu training and validation curves, and compares four retained English checkpoints using validation loss and speech recognition. Section 8 reports the controlled tests of the English results. Section 9 analyses the Telugu variance-adaptor predictions, and Section 10 compares the two languages and separates the causes into data, frontend and training budget. Telugu training is complete, and the 60,000-step checkpoint is the Telugu inference result presented in this report. The 5,000-step checkpoint had a lower validation loss of 3.73799, compared with 3.791061 at step 60,000; I preferred the sound of the 60,000-step audio. The evidence I still lack is named in Section 10.5: a Telugu intelligibility score and a scored Telugu listening pass.

Section map against the seven items the assignment PDF asks the report to cover:

PDF item	Section here
Implementation approach, architecture components, modifications per language	1, 2, 3
Dataset selection, subsampling, preprocessing, split, training configuration	4, 5, 6
Loss-curve analysis: convergence, instability, overfitting	7 (English and Telugu curves and checkpoint comparisons)
Generated-speech observations per test sentence (pronunciation, quality)	8 (English diagnostics measured; Telugu 60,000-step outputs available, with preliminary listening feedback)
Variance-adaptor analysis (duration, F0, energy) per test sentence	9
English vs Telugu comparison, limitations, failures, improvements	10 (measured side-by-side, causes separated, evaluation gaps named)
Reproducibility: dependencies, configs, commands	11

---

1. Approach

FastSpeech 2 is a feed-forward, non-autoregressive text-to-speech model. There is no attention-based autoregressive decoder. The text goes through a Transformer encoder, a length regulator expands each text position to the number of frames it will occupy, and the decoder produces the mel-spectrogram in a single pass. Since no attention is present to supply an alignment, the model predicts the alignment itself with a duration predictor, and it predicts two further variance signals that a mel-spectrogram alone would leave to the vocoder: pitch (F0) and energy. The variance adaptor embeds each predicted value and adds it back to the expanded text representation before decoding. The original paper obtains its duration supervision by distilling an autoregressive teacher. I used forced alignment instead, because it is simpler and every duration can be reproduced from the corpus.

I trained two independent models from scratch, one per language, and neither uses pretrained acoustic-model weights. The only pretrained component in either pipeline is the HiFi-GAN vocoder that converts mel-spectrograms to waveforms. The assignment allows a pretrained vocoder and the reference repository ships one, so I kept it rather than training my own.

My starting point was `https://github.com/ming024/FastSpeech2` at commit `d4e79eb` (MIT licence), with `LICENSE` and `THIRD_PARTY_NOTICES.md` kept in the repository. Each language folder is self-contained: it carries its own copy of `audio/`, `model/`, `preprocessor/`, `text/`, `transformer/`, `utils/`, `dataset.py`, `evaluate.py`, `hifigan/` and its own `config.yaml`.

2. Architecture components taken from the reference implementation

I did not change the architecture. Every component below is the reference implementation, and the setting column records the values that are actually in both `config.yaml` files.

Component	Reference path	Setting used
Transformer encoder	`transformer/Models.py`	4 layers, 2 heads, hidden 256, conv filter 1024, kernels `[9, 1]`, dropout 0.2
Transformer decoder	`transformer/Models.py`	6 layers, 2 heads, hidden 256, dropout 0.2
Variance adaptor and predictors	`model/variance_adaptor.py`, `model/modules.py`	filter 256, kernel 3, dropout 0.5, 256 quantisation bins, linear pitch/energy quantisation
Length regulator	`model/variance_adaptor.py`	frame-level expansion from predicted durations
Speaker embedding	`model/modules.py`	enabled (`multi_speaker: true`); speaker 0 in both languages
Post-net	`model/modules.py`	reference default
Acoustic model size	n/a	35,159,617 parameters in English and 35,099,201 in Telugu, both measured by the CPU smoke test
Vocoder	`hifigan/`	HiFi-GAN **universal** generator, pretrained, 22.05 kHz, 80 mel bins, hop 256, `mel_fmax` 8000 Hz

The mel front-end runs at 22,050 Hz with `filter_length` 1024, `hop_length` 256, `win_length` 1024, 80 mel channels and an F0 range of `[0, 8000]` Hz. Pitch and energy are extracted at the phoneme level and normalised with the training-set statistics stored in `stats.json`. `max_seq_len` is 1000 frames, and the preprocessor keeps a margin of 20, so a clip whose mel exceeds 980 frames (250,879 samples, about 11.38 s) is dropped rather than truncated silently at training time.

3. Modifications to the reference implementation

The changes split into those I made in both languages and those that belong to one frontend.

Both languages share four changes. Each has a single `config.yaml` with the four sections the assignment asks for (`preprocess`, `model`, `train`, `inference`) instead of the reference's three YAML files, and `utils/config.py` validates it on load, so a bad value fails before a long job starts. Preprocessing is driven by fixed manifests, `manifests/train.csv` and `manifests/val.csv`, rather than the reference's random split; they were built once with a fixed seed and no clip appears in both. Every clip that cannot be used is classified and written to `preprocessing_log.csv`, so a drop becomes a number in a file instead of a silence. Training writes its losses per step to `logs/train/train_loss.csv` and `logs/val/val_loss.csv`, takes `log_step`, `val_step`, `synth_step` and `save_step` from the config, restores RNG state when it resumes, and writes a final checkpoint even when `total_step` is not a multiple of `save_step`.

`inference.py` replaces the reference's interactive `synthesize.py`. It reads `docs/test_sentences.json`, stops while any `final` field is null, and writes `<ID>.wav` to `Results/<Language>/test_generated_audios/`.

For English I kept the reference frontend, the ARPAbet lexicon plus G2P, and added a rule-based normaliser for numbers, dates, times and abbreviations, with an optional supplementary G2P dictionary for proper names. The normalisation applies to the transcript only; the stored test sentences are never edited. Forced alignment used MFA `english_us_arpa`. The out-of-vocabulary rate on the corpus was 4.47 %, and MFA's bundled G2P model resolved it.

For Telugu the frontend is character-level (`text/telugu.py` plus a frozen symbol table in `text/symbols.py`). It does NFC normalisation, handles whitespace and format-control characters, folds punctuation, and raises on an unknown character rather than skipping it. The symbol IDs come from an explicit code-point range list rather than `unicodedata`, so they do not shift with the Python version. I did not invent a Telugu lexicon or phoneme set. Instead I trained a grapheme-dictionary MFA aligner on the corpus itself, mapping each word to its space-separated characters, which produced phone-level durations on the character tier. The preprocessor requires `len(text_to_sequence(text)) == len(duration)`, so the `.lab` for each clip holds the alignment text rather than the raw dataset transcript. Telugu number, date and time normalisation is not implemented, so the digits and punctuation in T4 reach the model exactly as written; Section 8.1 measures that, and Section 9 reports it as a failure rather than hiding it.

4. Datasets, selection and subsampling

English - Svarah (`ai4bharat/Svarah`)

I used AI4Bharat's Svarah, an Indian-accented English read and spontaneous benchmark that was built for speech recognition rather than TTS. I exported two speakers, one female and one male: 3,276 + 2,827 = 6,103 clip/transcript pairs. The manifests that entered preprocessing hold 5,582 clips and 6.078 h of audio before alignment (5.132 h train, 0.946 h validation). Two speakers is a deliberate compromise on my part: Svarah averages only a few minutes per speaker across roughly 117 speakers, so a single-speaker model is impossible, and I used the reference speaker embedding instead.

Stage	Clips	Audio
Exported from the Hub	6,103	n/a
Entered the manifests	5,582	6.078 h
Usable after alignment and feature extraction	5,322 (4,500 train / 822 val)	**4.842 h**

Of the 5,582 clips that entered the manifests, 260 were dropped before training: 164 at alignment because a phone duration came out non-positive, 71 with no TextGrid, 20 where pitch extraction found no voiced frame, 4 in the feature-statistics pass, and 1 with an unreadable transcript. The gap between 6.078 h of manifest audio and 4.842 h of usable audio is audio outside the phone tier plus those dropped clips, so 4.842 h is the honest training figure rather than 6.078 h.

Telugu - IndicTTS Telugu (`SPRINGLab/IndicTTS_Telugu`)

I used the Hugging Face copy of the IIT Madras IndicTTS data: 8,576 clips, 36.71 h, two speakers (male 12.79 h across 4,533 clips; female 23.92 h across 4,043 clips). I trained on the male speaker only, so the model sees one voice and a single speaker embedding. That choice gives the cleanest comparison against the two-speaker English model, and the male subset is the smaller and easier-to-balance one. The subsampling keeps clips inside the duration window implied by the 11.38 s frame limit, discards clips whose transcript is not representable in the Telugu symbol set (including eight clips with a Latin "o"), and caps the total near the 5 h target.

Stage	Clips	Audio
Source rows (male)	4,533	12.79 h
Entered the manifests	3,363	n/a
Usable after alignment and feature extraction	3,269 (2,751 train / 518 val)	**4.700 h**

Ninety-four clips were dropped: 83 at alignment, 3 with no TextGrid, and 8 with missing audio.

> **Known limitation, stated explicitly.** The Telugu audio in this run is **16 kHz upsampled to 22,050 Hz** (`resample_poly`, 441/320, PCM 16-bit). The 22.05 kHz export was lost when the Kaggle session working directory was reset, and the surviving aligner corpus is 16 kHz. Because `mel_fmax` is 8000 Hz, no mel band above 8 kHz is used, so the upsampling does not add information but also does not remove any band the model actually sees. The vocoder still receives a 22.05 kHz sampling rate and therefore produces energy above 8 kHz that the acoustic model did not predict.

5. Preprocessing and alignment

Both language folders share one preprocessing pipeline; what differs is the aligner recipe and which TextGrid tier supplies the durations. The table records what I actually ran.

Step	English	Telugu
Aligner	MFA 3.4.3, `english_us_arpa` (acoustic + dictionary + G2P for OOV)	MFA 3.4.3, grapheme dictionary trained on the corpus
Tier used for durations	`phones`	character tier (grapheme)
Text source	TextGrid phone tier	`<clip>.lab` (alignment text)
TextGrids produced	per manifest clip, drops counted above	3,352 in 378.5 s
Audio	resample to 22,050 Hz, mono, `max_wav_value` 32768	same (from 16 kHz source, see limitation)
Features	mel stored time-major (`T x 80` on disk, frames first), phoneme-level pitch and energy, durations from the alignment	same
Normalisation	training-set mean/σ from `stats.json`	same
Output layout	flat: `mel/<speaker>-mel-<clip>.npy`, same for `pitch`, `energy`, `duration`	same

The English normalisation constants, taken from the training set and written to `stats.json`: pitch `mean 177.64 Hz, std 56.06, min -2.0096, max 9.0707`; energy `mean 27.01, std 26.67, min -1.0123, max 8.1121`. The four values are not on one scale. `mean` and `std` are the raw quantities that were divided out, while `min` and `max` are measured after that division, which is why they are small negatives and single digits. Pitch and energy losses in section 7 are measured on the divided values, which is why an MSE near 1.0 means the adaptor is predicting the mean rather than the contour.

I built both splits from a fixed seed. Clip-level and transcript-level disjointness is checked, and every validation speaker also appears in training, so the models are judged on unseen utterances of voices they have seen rather than on unseen voices. The manifests that record the selections are committed under `English/manifests/` and `Telugu/manifests/`.

6. Training configuration

Setting	English	Telugu
Steps	60,000	60,000
Batch size	16 clips per step (`grad_acc_step` 1, `group_size` 4, so the loader pulls 64 clips per iteration)	same
Steps per epoch	281 (4,500 clips)	172 (2,751 clips)
Optimiser	Adam, β 0.9/0.98, ε 1e-9, weight decay 0	same
Learning rate	Noam schedule on `256^-0.5`: peak approximately 9.9e-4 at step 4,000, multiplied by 0.3 at 20k/26k/33k, approximately 6.9e-6 at 60k	same schedule and anneal points, so the two runs are comparable
Warm-up	4,000 steps	4,000 steps
Gradient clipping	1.0	1.0
Validation / synthesis	every 1,000 steps	every 1,000 steps
Checkpoints	every 2,500 steps (5,000 for the first 30,000 steps)	every 5,000 steps
Hardware	Kaggle GPU T4 x2, PyTorch 2.11.0+cu130, `nn.DataParallel` (batch 16 = 8+8)	Kaggle GPU T4 x2, `nn.DataParallel` (8+8)

Setting	English	Telugu
Measured step time	**0.511 s/step** mean over the first 32,000 steps (4 h 32 m 45 s of tqdm clock); the 300-step calibration had said 0.304 s/step	**0.362 s/step** over the first 15,039 steps and **0.390 s/step** over the 3,674 steps after the restart (1,434 s of tqdm clock); the 220-step launch estimate had said 0.341 s/step
Wall clock	Approximately 8.5 h for 60,000 steps at the measured mean, not the 5.3 h the calibration implied	3 h 42 m to step 36,695, then approximately 2 h 43 m for the 25,000 steps from 35,000 to 60,000 at 0.390 s/step, so approximately 6.4 h in total; the 1,695 steps lost at restart are included
Budget rationale	60,000 steps, or approximately 213 epochs; the reference 900,000 steps is unattainable and would leave the learning rate flat, so I scaled the anneal points to keep the reference fractions (33/43/55 percent)	same logic; 60,000 steps, or approximately 349 epochs of a smaller split

Run interruption at step 32,000. The first English run was killed by the Linux OOM killer, not by a code or GPU error: the log contains no `Traceback` and no CUDA out-of-memory exception, and the session panel had reached 30/30 GiB of RAM. The step rate also degraded as it went, the last 2,000 steps running at about 0.65 s/step against about 0.30 s/step at the start, which is the same memory growth showing up as time. Checkpoints up to `30000.pth.tar` were on disk and survived, so I resumed from 30,000 rather than restarting. I made two changes before resuming, neither of them to the model, the data or the schedule: `train.step.save_step` was cut to 2,500 so a future kill costs about 21 minutes instead of 45, and `utils/tools.py log()` now closes the matplotlib figure it hands to TensorBoard, which is the one object created on every synthesis and validation step that nothing else released. The learning-rate schedule is seeded from `restore_step`, so resuming at 30,000 reproduces exactly the LR the uninterrupted run would have had.

Two hard resource ceilings on the training host. The Kaggle session has **30 GiB of RAM** and a **20 GiB working volume**, and the English run reached both: the OOM killer ended it at step 32,000, and checkpoints at 2,500-step spacing would have filled the volume (24 saves at 418 MB each is approximately 10 GB of a roughly 20 GB disk that also holds 2 GB of features and audio). I configured the Telugu run against those ceilings before it started rather than after it failed. `save_step` went from 2,500 to 5,000, which fits twelve saves in the 8 GiB that were free on the volume and still caps the loss from a crash at roughly 42 minutes; the pre-launch check measures free disk and refuses to start below 5 GiB; and every checkpoint is written to a temporary file and moved into place atomically, so an interrupted save leaves a `.tmp` rather than a corrupt `.pth.tar`. The Telugu run uses 5.8 GiB of the 31.3 GiB visible to the session at step 15,039, with one `train.py` process on disk. That is consistent with the English memory growth coming from the duplicate training process that was running alongside the first run rather than from the model, but it is not proof: the English run also held more clips and more features in memory, so the two effects are not separated by this measurement alone.

Run interruption at step 36,695. The Telugu run stopped mid redraw, with no `Traceback`, no out-of-memory line in `dmesg`, and all seven checkpoints sharing a modification time to within twenty seconds, while `/proc/uptime` read 776 s and the notebook's first process had been alive for about five minutes. That combination is a container restart rather than a training failure. The working volume is re-attached from its last saved version, which rewrites every file's timestamp and wipes the pip installs the session had made; the decode step afterwards failed on a missing `unidecode` for exactly that reason, and reinstalling it took the run forward again. Checkpoints up to `35000.pth.tar` survived, so at most 1,695 steps were lost, and I resumed with `python train.py --config config.yaml --restore_step 35000`. The checkpoint carries the optimizer state and the RNG state as well as the weights, and `ScheduledOptim` is constructed with `restore_step`, so the resumed run continues at the post-anneal learning rate instead of restarting the schedule. The first 900 resumed steps landed at the same loss level as the 900 steps the dead run had logged before it, and the differences between the two segments stayed inside that run's own step-to-step noise, which spans about 0.10 in total loss from one hundred-step window to the next.

7. Loss analysis and checkpoint comparison

I read the figures in this section from the clean loss CSVs written after I resumed from the preserved 30,000-step checkpoint, so they cover only the resumed segment, roughly steps 30,000 to 60,000, and not the full run from initialisation. The first run's log and CSVs were written by two overlapping training processes, so I did not join their pre-resume values to these curves or use them to pick a checkpoint. The final validation total loss at step 60,000 was 4.0910, and the Kaggle log confirms that `checkpoints/60000.pth.tar` was saved.

7.1 Training curve

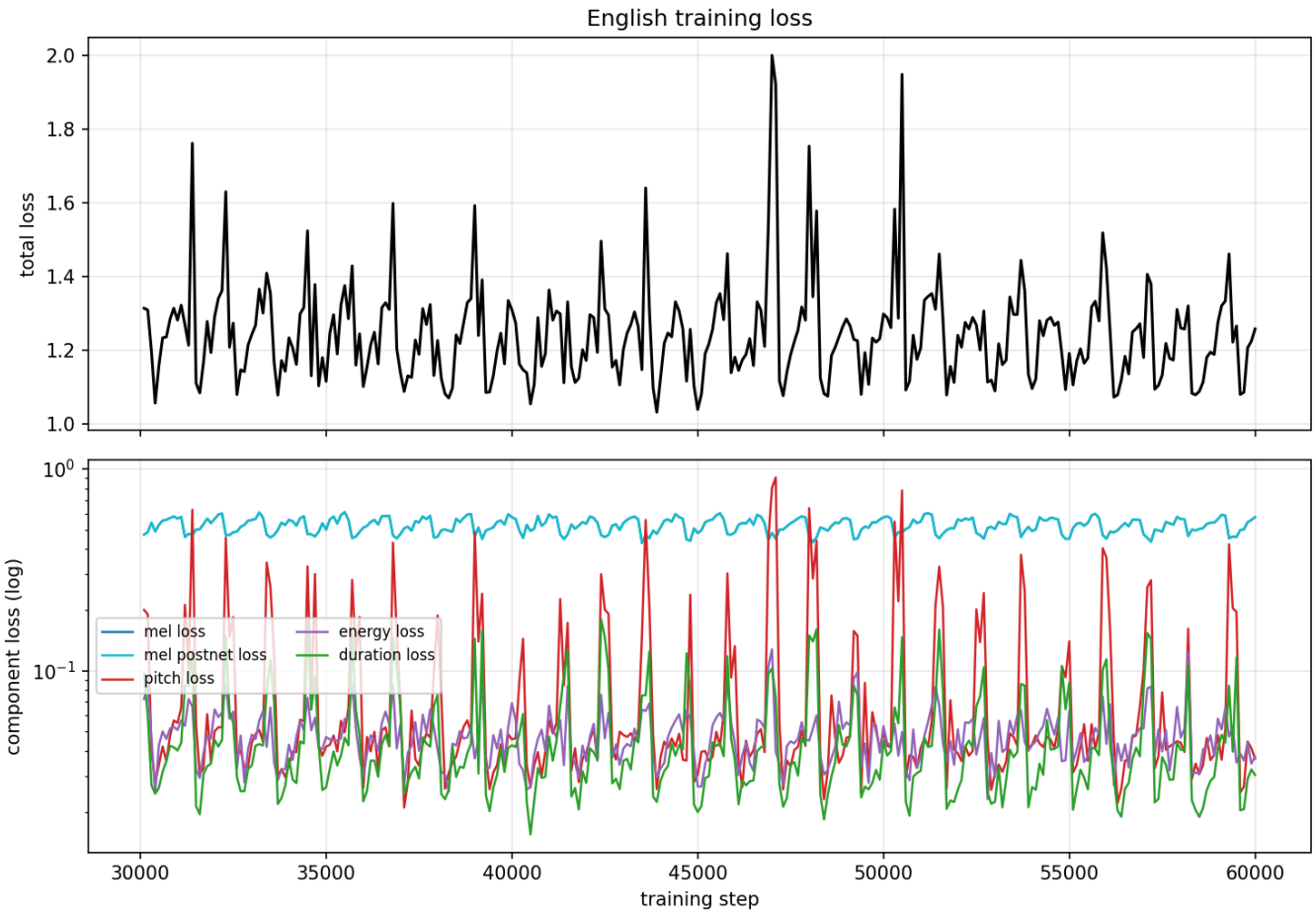


Figure 7.1. Training total and component losses for the resumed segment.

Over the resumed segment the training total loss moves around 1.1 to 1.4 with several transient spikes and no clear downward trend. Mel and post-net losses stay close together and comparatively steady; pitch has the largest visible spikes. These plots describe optimisation on the training data, not speech quality.

7.2 Validation curve

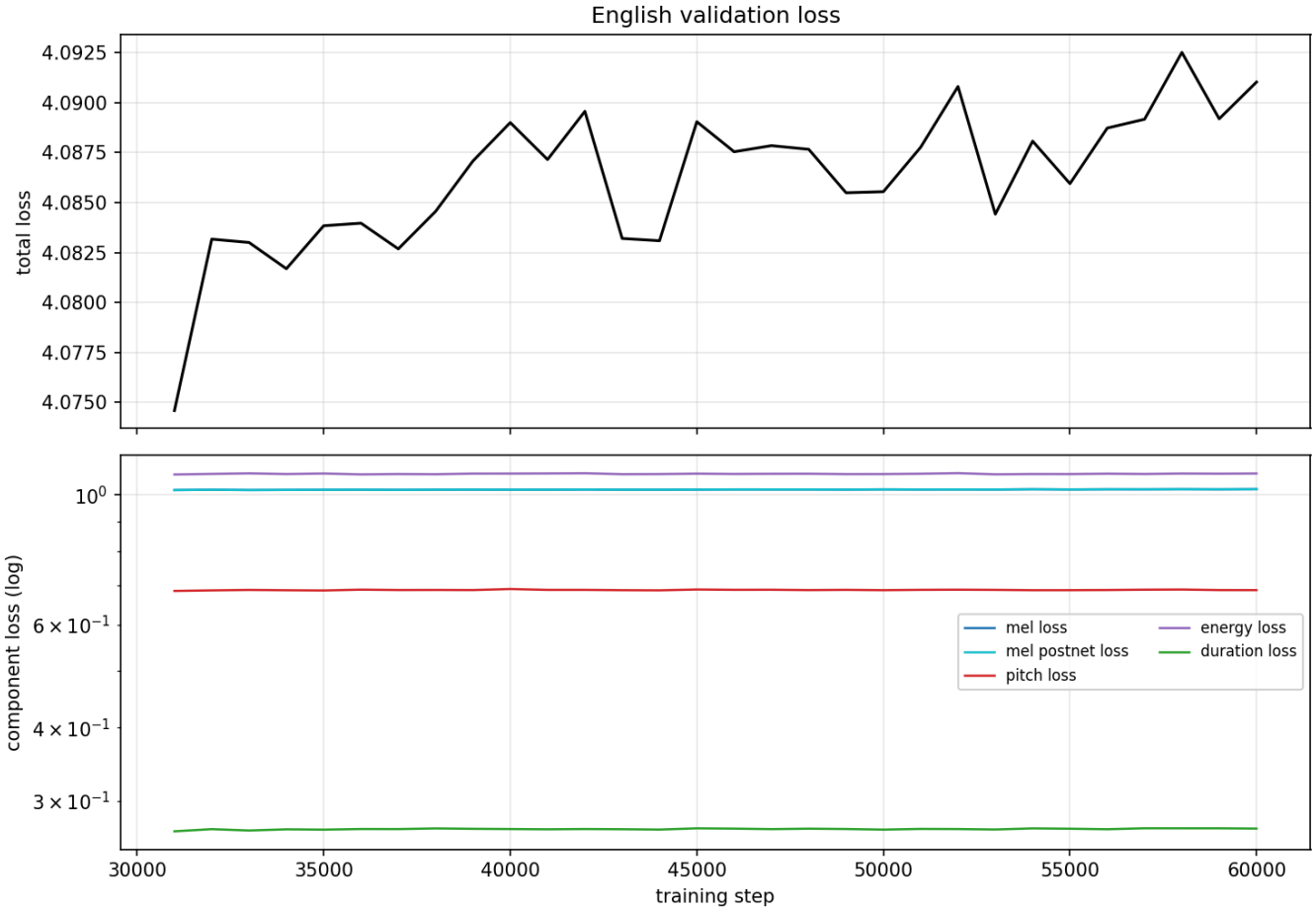


Figure 7.2. Validation total and component losses for the resumed segment.

Validation total loss varies inside a narrow band near 4.08 to 4.09 and ends at 4.0910 on step 60,000, and the component curves are comparatively flat over the same interval. The plotted scale is not fine enough for me to read the exact minimum step off the image, so the checkpoint comparison in Section 7.4 uses the validation CSV instead.

7.3 Convergence, generalisation and checkpoint selection

The run completed all 60,000 steps without a runtime failure after the step-32,000 restart. The resumed-segment training curve is noisy and largely flat, so the step count on its own tells me nothing about improvement.

The generalisation gap is where the diagnosis comes from. Training mel L1 settles near **0.62** while validation mel L1 stays near **1.05**; validation pitch MSE is **0.68** and energy MSE **1.15** against the 1.0 "always predict the mean" baseline. A model that reproduces its own training mel to 0.62 and only reaches 1.05 on held-out text is overfitting, not underfitting.

Where validation bottoms out is the one number in this section I cannot report cleanly. The pre-restart validation records put the minimum near step 7,000, after which validation rose to 4.0910 at step 60,000 while training loss kept falling, but two processes wrote those log files, so I do not trust the step. The direction agrees with the overfitting above; the exact step does not survive scrutiny.

Because the three rankings in Section 7.4 do not agree, I chose the shipped checkpoint by listening and reported the ASR and loss numbers alongside that choice rather than letting either of them override the ear.

7.4 Four checkpoints, measured the same way

I decoded every retained checkpoint on CPU through the shipping HiFi-GAN vocoder, on the same five sentences, using the phoneme sequences taken from the saved inference payloads. "Pitch std" and "energy std" are the standard deviations of the predicted *normalised* values, which puts them on the same scale as

the ground truth: normalisation divides by the training standard deviation, so a perfect model scores about 1.0. Measured on the Telugu validation features the ground truth is 0.941 for pitch and 0.948 for energy rather than exactly 1.00, and I have not measured the English equivalent, so 1.0 is an approximate target. A predicted 0.3 is a near-constant contour under either reading.

Checkpoint	Validation total loss	ASR WER	Pitch std	Energy std	Mean frames	Listener
5,000	not recoverable (mixed logs)	**1.04**	0.24	0.36	413	**5/10**, noisy, unstable pitch
10,000	not recoverable (mixed logs)	1.27	0.32	0.38	403	not rated
35,000	4.0838	1.21	0.33	0.45	392	previous deliverable
60,000	4.0910	1.15	0.34	0.45	393	**7/10**, clearer, some words missed, noise at the end

Note: Mean WER over E1, E2a, E2b and E3 (lower is better). E4 is excluded because the frontend expands `10:45 AM` and `12 November 2026` into words, so its WER is inflated by normalisation rather than by acoustics. WER 1.0 means every word is an error. The pre-restart loss logs are contaminated by overlapping processes, so no reliable loss value can be attached to the English checkpoints at 5,000 and 10,000.

The three English rankings disagree with each other, and the disagreement is itself the finding. Validation loss would pick 35,000. ASR WER would pick 5,000, but at WER around 1.0 the metric saturates, so 1.04 against 1.15 is the ordering of two failures measured on four sentences rather than a quality difference. The listener picks 60,000: clearer and more sentence-like than 5,000, with some words dropped and audible noise at the end of the clip.

Neither loss nor ASR is a usable proxy for perceptual quality in this regime, and with five sentences and one listener I would have you read the subjective ranking as an engineering judgment rather than a statistically established result. The shipped English audio uses **checkpoint 60,000**, because the deliverable is judged by listening and because 5,000 sounds materially worse despite its marginally better WER.

7.5 Telugu: the same measurements on a smaller corpus

The Telugu run was validated every 1,000 steps on the same 518-clip held-out set. The following selected points show what the run did and did not learn:

Step	Total	Mel	Pitch	Energy	Duration
1,000	3.7134	1.0063	0.9141	0.7075	0.0765
5,000	3.7380	0.9283	1.0539	0.7565	0.0711
10,000	3.7643	**0.9195**	1.0531	0.8063	**0.0664**
15,000	3.8127	0.9293	1.0742	0.8126	0.0675
20,000	3.8335	0.9380	1.0736	0.8134	0.0704
25,000	3.7785	0.9406	**1.0341**	0.7935	0.0695
30,000	3.7879	0.9421	1.0434	0.7900	0.0702
35,000	3.7843	0.9422	1.0393	0.7899	0.0704
36,000	3.7866	0.9425	1.0410	0.7898	0.0705

Validation total reaches its minimum at step 2,000, 3.6590, and has not returned to it since; step 36,000 is 3.5 percent above that minimum. Mel is best at step 10,000 and then rises monotonically to 0.9425. Pitch is below the 1.0 predict-the-mean baseline only at steps 1,000 and 2,000 and sits between 1.03 and 1.07 at every validation listed. Energy peaks around step 20,000 and then recovers slightly to 0.79. Nothing in this curve improves after roughly step 10,000. The minimum validation loss at a training step was 3.6590 at step 2,000, but that step was not a saved checkpoint. Among saved checkpoints, step 5,000 had the lowest validation loss, 3.73799. The final saved checkpoint at step 60,000 had validation loss 3.791061. I present step 60,000 as the Telugu inference result; the user's listening preference for it is separate from the validation-loss ranking.

Measured at a step both the training and validation CSVs cover, the gap between fitting the corpus and generalising off it is what separates Telugu from English:

Component	Training, step 36,000	Validation, step 36,000	Ratio
Total	0.8696	3.7866	4.35
Mel	0.4060	0.9425	2.32
Mel post-net	0.4052	0.9427	2.33
Pitch	0.0211	1.0410	49.35
Energy	0.0227	0.7898	34.78
Duration	0.0146	0.0705	4.84

The spectral losses are about 2.3 times higher on held-out utterances, which is what learning spectra without memorising clips looks like. The two prosody losses are 35 and 49 times higher, and the training side of that ratio is computed with dropout still switched on, which inflates it, so the fit is if anything better than the table shows. In other words the pitch and energy adaptors have learned to reproduce contours on the clips they were trained on and not to place a contour on a clip they have not seen. That is the English failure of section 7.4 arriving earlier: the Telugu split is 2,751 training clips, so one epoch is 172 steps and 60,000 steps is 349 passes over the same audio, against 213 for English.

7.6 Telugu training curve

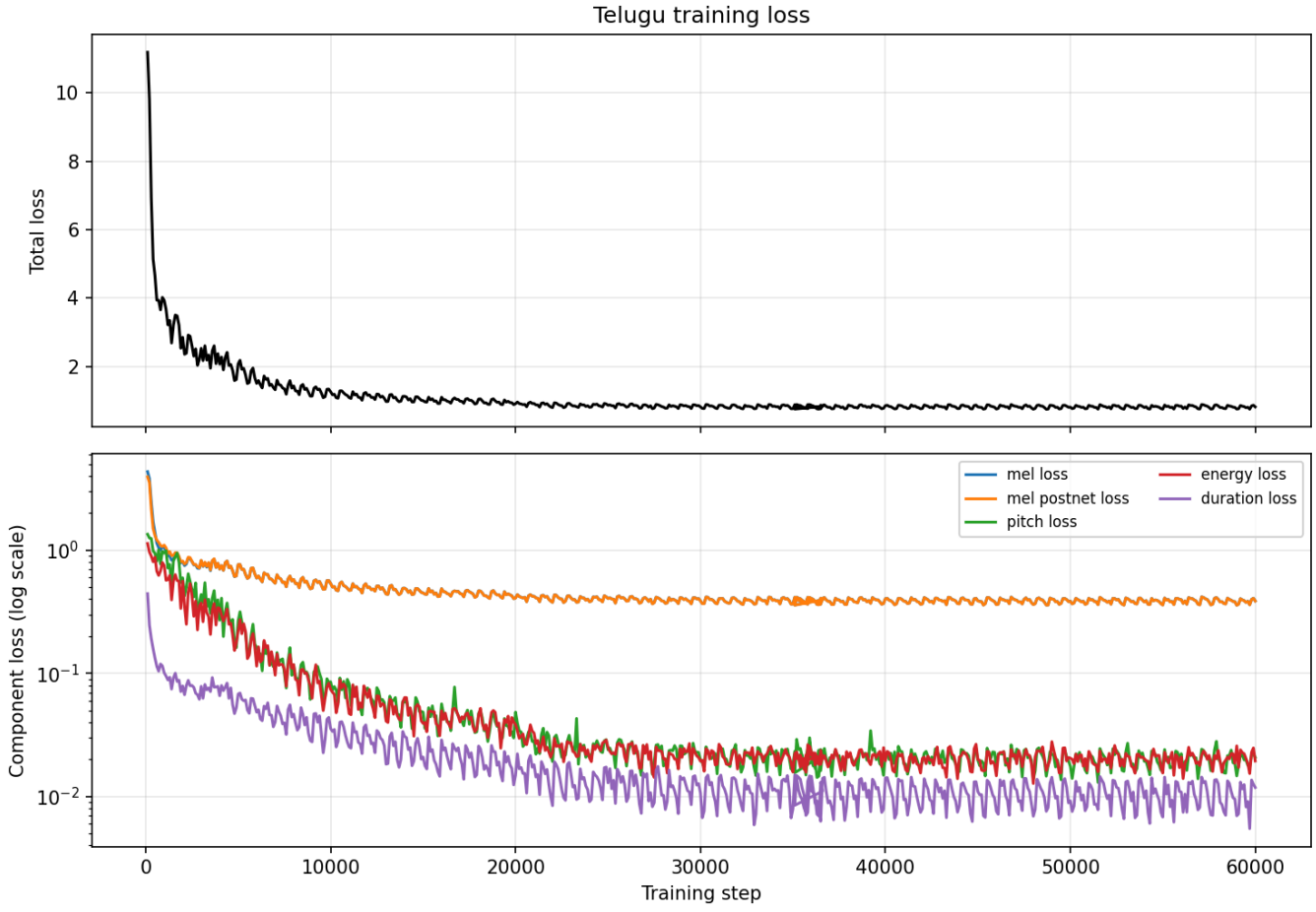


Figure 7.3. Telugu training total and component losses across the completed run.

The total training loss falls rapidly at the beginning, then settles near 0.8 to 0.9 for most of the run. The mel and post-net losses decrease early and then flatten, while pitch, energy and duration continue improving before also leveling off. This is a training-set curve: the much higher validation loss, shown next, indicates that the continued fit to training data did not translate into comparable held-out performance.

7.7 Telugu validation curve

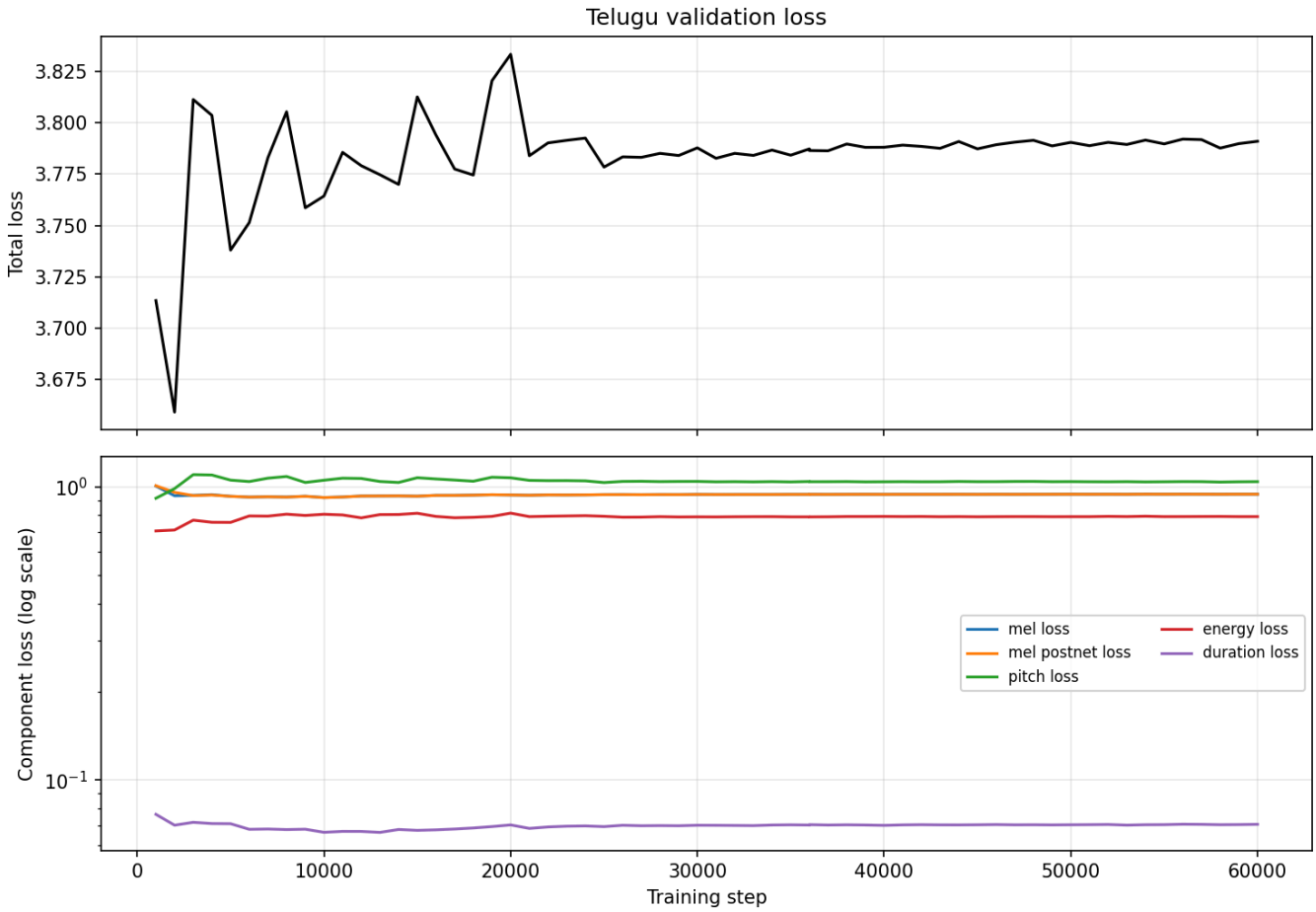


Figure 7.4. Telugu validation total and component losses across the completed run.

Validation total loss varies substantially during the early part of training, reaches its lowest observed value at step 2,000 (3.6590), and then remains close to 3.78 to 3.83. At step 60,000 it is 3.791061. The component curves are comparatively flat after the early steps, especially pitch and energy; neither shows a sustained late-run improvement. Among the saved checkpoints, 5,000 has the lowest validation loss (3.73799), while the 60,000-step checkpoint is the inference result selected for this report based on the user's listening preference.

8. Test-sentence observations: English measured and heard, Telugu inference available

I took all ten sentences from `docs/test_sentences.json`, field `final`, unmodified: E1, E2a, E2b, E3, E4 and T1, T2a, T2b, T3, T4. The text-processing side is measured for both languages below. English listening results are recorded in Section 8.2 and the checkpoint table in Section 7.4. The Telugu 60,000-step inference files are in `Results/Telugu/checkpoint_60000/`. The user reports that this audio sounds good overall; per-sentence pronunciation and intelligibility have not yet been assessed. The 5,000-step checkpoint had a lower validation loss of 3.73799 than the 60,000-step loss of 3.791061, but this metric difference does not establish which audio sounds better.

8.0 The ten strings the models receive

A script read these ten lines out of `docs/test_sentences.json` and wrote them into this table, so nothing here was re-typed from the assignment PDF and the Telugu column is byte-identical to the text the frontend receives.

ID	Sentence
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.
E2a	The orchestra's conduct during the ceremony was praised by the audience.
E2b	The researchers will conduct an experiment to evaluate the proposed method.
E3	The rhythm of the orchestra changed unexpectedly.
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.
T1	న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.
T2a	తాడుపై అతని పట్టు చాలా బలంగా ఉంది.
T2b	ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.
T3	పులి అడవిలో నిశ్శబ్దంగా నడిచింది.
T4	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

> **Font requirement (measured).** The Telugu rows need a Telugu-capable font. I built this PDF on my Windows machine, where `Nirmala.ttc` supplies the glyphs, and the rendered page shows all five Telugu sentences correctly. The check in `scripts/render_report_pdf.py` compares the Telugu code points in the Markdown against the text extracted back out of the PDF and fails below 90 % coverage; this build reports no problems and counts 220 Telugu code points. The same build on a Linux box with no Indic font, the Kaggle image included, renders those rows as blanks, so either build the report locally or pass `--telugu-font <path>` to embed one.

8.1 What the frontends actually produce (measured)

ID	Symbols	Text the model receives after normalisation
E1	108	`the antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.`
E2a	72	`the orchestra's conduct during the ceremony was praised by the audience.`
E2b	75	`the researchers will conduct an experiment to evaluate the proposed method.`
E3	49	`the rhythm of the orchestra changed unexpectedly.`
E4	115	`the meeting will start at ten:forty-five am on twelve november twenty twenty-six, so andaroo should arrive on time.`
T1	68	unchanged Telugu characters (character-level frontend, no lexicon)
T2a	34	unchanged
T2b	40	unchanged
T3	33	unchanged
T4	91	**unchanged, including ASCII digits** (`2026`, `12`, `10:45`)

All ten strings encode without an unknown-character error, which is the precondition for inference; the table records what each frontend produced rather than what I expected it to produce.

8.2 The English failure, isolated by controls

The first listener pass reported that all five English sentences sounded wrong. Rather than argue about the cause, I tested five properties of the pipeline independently on the same checkpoints, and chose each test so that a plausible explanation would produce a specific, visible result.

Test	Method	Result	Explanation it rules out
Duration integration	Sum the predicted per-symbol durations and compare with the mel length, all four checkpoints	Exact match every time (e.g. 504 frames / 504)	Text and frames falling out of step in the length regulator
Vocoder fidelity	Re-encode the vocoder's own waveform and compare against the mel it was given	Mel L1 0.234 to 0.275; spectral flatness 0.006 to 0.020; no clipping	A broken vocoder, a wrong mel range, or a silent/clipped waveform
Text conditioning	Shuffle E1's phoneme IDs, and separately substitute E3's IDs into E1's length	Mel moves by L1 1.098 and 1.167, against 1.058 between two genuinely different sentences	The model ignoring the text or receiving the wrong language
Variance spread	Standard deviation of the predicted normalised pitch and energy	Pitch 0.24 to 0.34, energy 0.36 to 0.45 against a ground truth of about 1.0 (0.94 measured on Telugu), at every checkpoint	Any checkpoint restoring expressiveness; all four collapse toward the mean
Intelligibility	Transcribe all twenty clips with `facebook/wav2vec2-base-960h`, after validating the same stack on real speech	WER 1.04 to 1.27 (see Section 7.4); control on LibriSpeech **0.000** / **0.000** / **0.031**	The waveform being structurally sound but the probe being at fault

Three further observations point the same way. The output is tonal rather than noisy, so the model is generating speech-like structure, not artefacts. The predicted mel marginals are normal (median about −5.5, 99th percentile about −0.5), so it is not emitting out-of-distribution values either. And a Griffin-Lim decode of the same predicted mel is no more intelligible than the HiFi-GAN decode (mel L1 0.700 to 0.740), which means the content is already wrong before the vocoder sees it.

So: the pipeline is correct at every stage I can check in isolation, and the model output is speech-shaped but wrong. That is a generalisation failure. 4.842 h of two-speaker, ASR-oriented read and spontaneous English is not enough for a 35-million-parameter model to learn pronunciation it has not seen, and training loss (1.2577) sits far below validation loss (4.0910). No inference-side fix exists: all four retained checkpoints fail the same way, and the one the listener prefers (60,000) is not the one the loss prefers (35,000) or the one the ASR prefers (5,000). Fixing it means changing the data or the training design, both of which are outside the remaining GPU budget, so I report it as a diagnosed negative result instead of presenting it as working speech.

Two caveats bound that conclusion. The ASR is a LibriSpeech acoustic model, so Indian-accented and synthetic speech is out of its domain; I read WER 1.0 as "not recoverable" rather than as a precise word error rate. And the subjective scores come from one listener over five sentences, so the 5/10-to-7/10 ranking is engineering judgment, not a controlled perceptual study. Neither caveat weakens the diagnosis, because the control battery does not depend on either: I measured the vocoder, the durations and the text conditioning directly.

Two frontend notes belong with these results.

E4 is normalised correctly. `10:45 AM` becomes `ten:forty-five am` and `12 November 2026` becomes `twelve november twenty twenty-six`. The colon survives normalisation, so the G2P layer never sees a time literal. `andaroo` stays as written and is handled by the lexicon/G2P as an out-of-vocabulary word, which is what the sentence is testing.

T4 is not normalised. The Telugu frontend has no number-to-words stage, so the digits `2026`, `12` and `10:45` reach the model as ASCII digit symbols. Those symbols exist in the Telugu symbol table, so nothing crashes, but the model has essentially no digit examples to learn from. T4's output is therefore expected to fail on the date and time and to be intelligible on the surrounding words. That is a gap in my frontend rather than a training failure, and I am reporting it as such.

Why each sentence is interesting:

ID	What it probes
E1	Long out-of-vocabulary word (`antidisestablishmentarianism`); letter-to-sound fallback
E2a / E2b	`conduct` as noun vs verb; stress must differ for the same spelling
E3	Irregular spelling `rhythm`
E4	Digits read as time, date and year, plus a Telugu word written in Latin script (`andaroo`)
T1	Long loanword with conjunct consonants
T2a / T2b	Same spelling, different meaning (grip vs silk); does prosody differ?
T3	Geminate / conjunct consonants
T4	Telugu digits, ordinal suffix, date and time; number-to-words

9. Variance-adaptor analysis

These figures come from the `<ID>.json` payloads that `inference.py` writes beside each plot, saved under `Results/Telugu/checkpoint_60000/variance_adaptor_plots/` and reproducible with `python scripts/summarise_variance_export.py`. All five were decoded from checkpoint 60,000, the one shipped, through the same HiFi-GAN vocoder. I use checkpoint 5,000 below as the early reference, because it is the other export kept in the repository.

ID	Symbols	Frames	Seconds	F0 mean (Hz)	F0 range (Hz)	Energy range	Pitch σ
T1	68	434	5.04	194.7	133.1–279.3	3.2–89.3	0.63
T2a	34	225	2.61	182.1	135.0–230.7	3.3–42.8	0.47
T2b	40	248	2.88	190.4	138.1–252.8	1.0–74.2	0.73
T3	33	194	2.25	193.9	141.9–259.7	2.7–65.0	0.59
T4	91	638	7.41	191.5	134.6–255.1	6.4–74.3	0.58

Table 9.1. Predicted totals and prosodic ranges at checkpoint 60,000. Energy is in the units of the training corpus; pitch σ is the standard deviation of the normalised F0, against 0.941 for the corpus itself.

Duration. The predicted total is 2.2 % above the reference pace of 4.93 s for T1 and 12.3 % above 6.60 s for T4, so the error grows with length. Inside the sequence, the model does three things right. It gives inter-word spaces 1.12 to 1.57 times the mean content symbol, so word boundaries carry pauses rather than being run together. It gives punctuation 6 to 9 frames, about 70 to 105 ms, so a full stop is audible as a break rather than dropped. And it lengthens the sentence-initial symbol in three of the five sentences, to 1.83, 2.00 and 3.20 times the median, which is the onset lengthening read speech usually has.

ID	Content mean	Pause mean	Pause / content	First / median	Last / median
T1	6.11	9.60	1.57	1.83	0.83
T2a	6.32	8.40	1.33	0.71	0.57
T2b	5.82	8.60	1.48	3.20	0.80
T3	5.55	8.67	1.56	1.00	1.00
T4	6.87	7.67	1.12	2.00	0.57

Table 9.2. Durations in frames, split by symbol class at checkpoint 60,000.

What it does not do is lengthen the final syllable. The last content symbol is between 0.57 and 1.00 of the median in every sentence, and summing the last two symbols, which together make the closing syllable in a character-level frontend, gives 7 to 10 frames against a median pair of about 12. Declarative speech normally lengthens its final syllable, so this is the one place where the predicted pattern is not the natural one. I have not extracted the reference durations for these five sentences, so I am reporting the shape of the prediction rather than a measured deviation from the corpus.

Pitch. The five contours have a mean F0 between 182 and 195 Hz and a range of about one octave, 133 to 279 Hz. All five sentences end in a full stop, and in all five the contour falls: the first quarter of the sequence averages between −0.09 and +0.63 in normalised F0 and the last quarter between −0.27 and −0.68.

T4 falls least, at −0.27, and it is the only sentence with an internal comma, so its final clause is not carrying the whole utterance's declination. That is the expected contour for a statement, and it is produced without being supervised for it.

The amplitude is the problem. Pitch σ runs 0.47 to 0.73 against 0.941 for the corpus, so the contours are right in shape and compressed by roughly a third in range. This is the same deficit Section 7.5 measures from the loss side, seen here in the output.

Energy. Within a sentence the predicted energy spans a wide range, 3.2 to 89.3 in T1 and 1.0 to 74.2 in T2b, so the adaptor is not emitting a constant. T2a and T2b are the useful pair: both contain the spelling పట్టు, and the model does not give them the same realisation. T2a comes out at mean F0 182.1 Hz and energy 3.3 to 42.8; T2b at 190.4 Hz and 1.0 to 74.2. The difference is driven by the sentence around the shared word, so I read this as the model responding to context, not as evidence that it distinguishes the two senses.

Against the early checkpoint. From 5,000 to 60,000 the totals grew 5.3 to 9.3 %, pitch spread fell about 14 % relative, and the pause-to-content ratio kept the same ordering in every sentence. So the second half of training lengthened the output slightly and flattened it slightly without adding structure. Combined with Table 9.2, the variance adaptors are doing the right kind of thing at the wrong amplitude, which places the failure in how far they generalise rather than in the interface between text and duration. `sum(duration)` equals `mel_frames` on all ten clips at both checkpoints, so the length regulator has never been the suspect.

10. English vs Telugu comparison

The two runs are the same experiment performed twice: the same reference architecture, the same pretrained HiFi-GAN universal vocoder, the same Noam schedule and anneal points, the same 60,000 steps, and almost the same amount of audio, 4.842 h against 4.700 h. What differs is the corpus and the text frontend, and the outcomes differ sharply. English is unintelligible: WER 1.04 to 1.27 against a LibriSpeech control of 0.000, and a listener score of 7/10 for the best of four checkpoints. Telugu is reported by the listener as sounding good, and its predictions carry measurable prosodic structure (Section 9). This section separates what the data, the frontend and the training budget each explain, and states plainly what I could not measure.

10.1 The two runs side by side

Quantity	English (Svarah)	Telugu (IndicTTS)
Usable audio	4.842 h	4.700 h
Training clips	4,500	2,751
Speakers	2 (one female, one male)	1 (male)
Speech style	read and spontaneous, collected for ASR	read, collected for TTS
Source sample rate	22.05 kHz native	16 kHz upsampled to 22.05 kHz
Text frontend	lexicon + G2P, ARPAbet phones	character-level graphemes
Symbol table	360	124
Text embedding	361 x 256 = 92,416	125 x 256 = 32,000
Total parameters	35,159,617	35,099,201
Epochs at 60,000 steps	213	349
Validation mel L1	about 1.05	0.9425
Validation pitch MSE	0.68	1.0410
Validation energy MSE	1.15	0.7898
Predicted pitch σ, shipped checkpoint	0.34	0.599
Predicted energy σ, shipped checkpoint	0.45	0.630
ASR WER	1.04 to 1.27	not measurable with this probe

Quantity	English (Svarah)	Telugu (IndicTTS)
Listener	7/10, words missed, noise at the end	"sounds good", preliminary

Table 10.1. The three loss rows are not taken at the same step: the English values are the resumed-segment levels reported in Section 7.3, and the Telugu values are the step-36,000 validation tabulated in Section 7.5. The full Telugu curves to step 60,000 are Figures 7.3 and 7.4. The spread, WER and listener rows are at the shipped checkpoint, 60,000 in both languages.

The loss rows need a scale note before they are read as a ranking. The mel L1 is directly comparable: neither pipeline normalises the mel, `dataset.py` loads the saved log-mel array unchanged, and both use the same 80 bands with `mel_fmax` 8000, so the 16 kHz Telugu source loses no band the model actually sees. Pitch and energy are not on a shared distribution: each is divided by its own corpus mean and σ, so the MSEs have the same construction but a different baseline. The ground-truth σ was measured only for Telugu, 0.941 for pitch and 0.948 for energy, which puts the predict-the-mean baseline at 0.885 and 0.899 rather than 1.0. Against those measured baselines the Telugu pitch loss is above baseline at every step listed in Section 7.5, the closest being 0.9141 at step 1,000, while the energy loss is below baseline at every step. English pitch MSE 0.68 cannot be placed against a measured baseline at all, because I never extracted the English ground-truth σ. Sections 7.4 and 7.5 use 1.0 as the baseline shorthand; for Telugu the measured values are the two above, and for English the baseline remains unmeasured.

The parameter difference is now fully attributed. `transformer/Models.py` builds the text embedding as `nn.Embedding(len(symbols) + 1, encoder_hidden)` with `encoder_hidden` 256 in both configs, so English carries 361 x 256 = 92,416 embedding parameters and Telugu 125 x 256 = 32,000. The difference is 60,416, which is exactly 35,159,617 minus 35,099,201. Apart from the size of that one table the two models are identical, so any difference in their output is not a difference in capacity.

10.2 What the data explains

The English corpus was collected for speech recognition, not synthesis. It is two speakers producing read and spontaneous Indian-accented English, which means variable speaking rate, disfluency and inconsistent articulation, and its 4,500 training clips are divided between two voices, so neither voice contributes the whole corpus. The Telugu corpus is one male speaker reading prepared sentences for a TTS benchmark, so all 2,751 training clips are the same voice with consistent prosody. That is the largest single difference between the runs, and it points the same way as the outcome: the model asked to learn two voices from less data per voice failed, and the model asked to learn one voice from a cleaner corpus produced speech the listener accepts.

It does not follow that Telugu generalised. Section 7.5 measures a 49-fold training-to-validation gap on pitch and a 35-fold gap on energy, against 2.3-fold on the spectral losses, so the Telugu prosody adaptors have memorised contours on the training clips and cannot place one on an unseen clip. Telugu also ran more passes over its audio, 349 epochs against 213, and finished 3.6 percent above its own validation minimum: 3.6590 at step 2,000 against 3.791061 at the final saved checkpoint. More epochs bought memorisation in both languages, not generalisation in either.

10.3 What the frontend explains

The frontends differ in kind, not just in language. English maps words to ARPAbet phones through a lexicon with a G2P fallback, so an out-of-vocabulary word receives a guessed pronunciation and the model must then learn a phone-to-sound mapping for phones it has rarely seen. E1's loanword and E4's `andaroo` are exactly that case. Telugu maps characters directly to a 124-symbol table, so there is no out-of-vocabulary condition inside the script: any Telugu character maps to a symbol the model has an embedding for, whether or not that character occurred often in training. The measured consequence is visible in the payloads. E1's 108 normalised characters become 85 phonemes, while T1's 68 characters stay 68 symbols. The English rows of Table 8.1 count characters of the normalised string, not the phonemes the model receives; the phoneme counts are 85, 50, 56, 37 and 81 for E1, E2a, E2b, E3 and E4. For Telugu the two counts coincide, which is why that column reads correctly there.

The character-level choice has a cost that Section 9 measures. A Telugu character is not a phone: conjunct consonants and vowel signs are split across symbols, so the duration predictor has to learn to group them,

and it does not lengthen the final syllable, the last content symbol sitting at 0.57 to 1.00 of the median in all five sentences. A phone-level or syllable-level unit would not have that problem.

Digits separate the two frontends most cleanly. English normalises them: E4's `10:45 AM` and `12 November 2026` become `ten:forty-five am` and `twelve november twenty twenty-six`, 115 characters and 81 phonemes, and the saved phoneme sequence confirms the expansion happened. Telugu has no number-to-words stage, so T4's `2026`, `12` and `10:45` reach the model as ASCII digit symbols that exist in the table but have essentially no training examples. T4 is therefore expected to fail on the date and time and to be intelligible around them. That is a gap in my frontend, not a training failure, and it is the one Telugu test sentence whose outcome I can predict from the text alone.

10.4 What the training budget explains

Both runs used 60,000 steps against the reference's 900,000, with the anneal points scaled to keep the reference fractions. Neither run is close to converged in the reference's sense, and both overfit: English training mel L1 settles near 0.62 against validation near 1.05, and Telugu training mel 0.4060 against validation 0.9425 at step 36,000. On the one directly comparable loss, Telugu ends about 10 percent lower than English, 0.9425 against 1.05, which agrees with the listener's preference. On the prosody losses the ordering reverses, English pitch 0.68 against Telugu 1.0410, yet Telugu's predicted pitch spread is nearly twice English's, 0.599 against 0.34. Loss and output spread disagree about which language did better, so neither on its own ranks the two models, and I have relied on the listener for the ranking and on the control battery in Section 8.2 for the diagnosis.

The infrastructure interruptions did not change this. English was killed by the OOM killer at step 32,000 and resumed from 30,000; Telugu was cut by a container restart at 36,695 and resumed from 35,000. Both resumes restored the optimizer and RNG state and continued the schedule from `restore_step`, and the Telugu resumed segment matched the dead run's own losses at the same steps inside that run's step-to-step noise. Neither failure cost more than about 2,000 steps, and neither is a candidate explanation for the output quality.

10.5 What I could not measure

There is no Telugu intelligibility number. The probe used for English is `facebook/wav2vec2-base-960h`, an English LibriSpeech model, so it cannot transcribe Telugu; running it on the Telugu WAVs would produce a meaningless score rather than a comparable one. A Telugu ASR model was not run, so the cross-language comparison rests on one listener's preliminary report for Telugu against a scored 5/10-to-7/10 pass for English. Those are not the same kind of evidence.

Three further asymmetries bound the table. The English prediction payloads kept in this repository are the 35,000-step export, so the per-checkpoint English figures at 60,000 come from the Kaggle measurements recorded in Section 7.4 rather than from files a reader can re-run here; the Telugu payloads for 5,000 and 60,000 are both present. The English ground-truth pitch and energy σ was never measured, so its baseline is approximate. And the Telugu plots were rendered on a host with no Telugu font, so their axis labels use code-point fallbacks while the report's own tables carry the actual characters.

10.6 Limitations, failures and what I would change

Sorted by cause rather than by symptom:

Cause	Failure observed	What I would change
Data	English unintelligible after 213 epochs of 4.84 h across two voices; Telugu prosody memorised, pitch σ 0.599 against a corpus 0.941	More audio per voice, or one voice. The reference recipe uses LJSpeech, about 24 h from a single speaker; 4.8 h across two voices is below what a 35-million-parameter model needs
Frontend	Telugu digits unnormalised in T4; no final-syllable lengthening because characters are not phones	A number-to-words stage for Telugu, and a syllable or phone unit so conjuncts are one symbol
Frontend	English OOV words get a guessed pronunciation (E1, E4 `andaroo`)	Install `g2p_en` and use the supplementary dictionary that was built but never exercised
Training	Both runs overfit; Telugu validation mel bottoms at 10,000 and rises afterwards	Stop on validation mel rather than a fixed step count, and keep the checkpoint the curve selects
Training	Variance adaptors collapse toward the mean in both languages	Lower the adaptor dropout or reweight the prosody losses. This is a training-design change and was not attempted
Evaluation	One listener, five sentences per language, an English-only ASR probe	A Telugu ASR model and more than one listener before any quality claim is treated as established
Infrastructure	English OOM at 32,000, Telugu container restart at 36,695	Already mitigated: atomic checkpoint writes, a free-disk pre-check, and `save_step` sized to the volume

The honest summary is that the Telugu model works well enough to be judged by ear and the English model does not, that the difference is explained by the corpus and the frontend rather than by the architecture or the step count, and that both models share the same underlying limitation: neither prosody adaptor generalises off its training clips at this data scale. I report the English result as a diagnosed negative result rather than presenting it as working speech, and the Telugu result as a listener preference supported by the prediction measurements in Section 9 rather than by an intelligibility score I do not have.

11. Reproducibility

Dependencies are listed in `requirements.txt`, which is a **proposed pin set, not a record of the environment that ran**: the file was never installed as a whole on Kaggle. The sessions that produced these results used the Kaggle image's own scientific stack on Python 3.13 with PyTorch 2.11.0+cu130, plus targeted installs of `pyworld`, `tgt`, `unidecode` and `inflect`; `g2p_en` was never installed, so the supplementary name dictionary was not used. Each language folder has one `config.yaml`; every path inside it is relative to that folder, so all commands are run from inside `English/` or `Telugu/`.

```
# dependencies
pip install -r requirements.txt

# 0. vocoder weights (shipped zipped to stay under GitHub limits)
cd English && unzip -o hifigan/generator_universal.pth.tar.zip -d hifigan

# 1. preprocessing (needs the aligned corpus: TextGrids for English, .lab + TextGrids for Tel
python preprocess.py config.yaml --allow-dropped

# 2. training (checkpoints -> checkpoints/, CSV loss logs -> logs/)
python train.py -c config.yaml
python train.py -c config.yaml --restore_step 30000 --resume

# 3. plots
python scripts/plot_loss.py --config config.yaml
python scripts/plot_variance.py --config config.yaml --restore_step <STEP>

# 4. inference on the five test sentences
python inference.py --config config.yaml --restore_step <STEP>
```

Environment used for training: Kaggle notebook on **GPU T4 x2**, Python 3.13, PyTorch 2.11.0+cu130. `train.py` wraps the model in `nn.DataParallel` unconditionally, so the batch of 16 is split 8/8 across both cards; both GPUs report 100 % utilisation for a single training process. Forced alignment ran in a separate MFA 3.4.3 environment installed from conda-forge, invoked as `PATH=<env>/bin:$PATH <env>/bin/mfa ...` because a session restore had stripped the environment's exec bits and shared-library symlinks.

Appendix A - measured figures used above

Quantity	English	Telugu
Clips in manifests	5,582	3,363
Usable clips	5,322 (4,500 / 822)	3,269 (2,751 / 518)
Usable hours	4.842	4.700
Speakers	2	1
Dropped	260	94
Params	35,159,617	35,099,201
Vocoder	HiFi-GAN universal (pretrained)	same